8-1. Generalised Linear Models (GLMs)
Introduction

- The **normal linear regression** model for a response variable Y_i on a single explanatory variable X_i is described by

$$\mathrm{E}\left[Y_i \mid X_i\right] \quad = \quad \mu_i, \quad i = 1, \ldots, n$$

with

$$\mu_i \quad = \quad \beta_0 + \beta_1 x_i, \quad Y_i \sim N\left(\mu_i, \sigma^2\right)$$

- Generalized Linear Model extends this idea to cases where:
  - the **normal** distribution is **not appropriate**
  - and/or the **identity relationship** between the linear function of the β parameters and μ_i **does not hold**.

Example 8.1

The table below shows numbers of stressful events reported by a sample of people for the previous 18 months, classified by 'date' = number of months before interview.

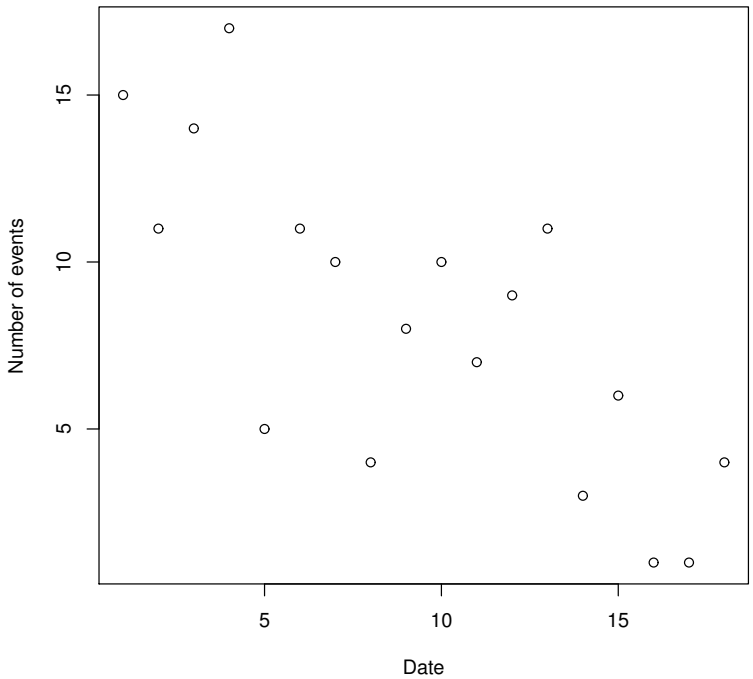
Date i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
Events N_i	15	11	14	17	5	11	10	4	8	10	7	9	11	3	6	1	1	4	147

Here the response variable N_i is a non-negative integer, so the simplest model is

$$N_i \sim \text{Poisson}(\mu_i)$$

where the N_i are assumed independent.

Plot the response variable N_i against the explanatory variable i:



It can be seen that the data show a decreasing trend. If we fitted a linear model, then extrapolating to the right could lead us to predict negative numbers of events. Instead, we can fit a (decreasing) exponential curve, called a *log-linear trend model*.

Log-linear trend model

Consider a 'trend model' in which N_x is the number of events in time period numbered x (or for a group indexed by x in some other circumstances) and

$$N_x \quad \sim \quad \text{Poisson}\,(\mu_x)$$

where the N_x are independent.

- Note that $E\,[N_x] = \mu_x$
- Suppose that these means have a trend described as

$$E\,[N_x] \quad = \quad \mu_x \quad = \quad \exp(\beta_0 + \beta_1 x)$$

  and note that this ensures that μ_x is non-negative.
- Can write the above relationship as

$$\log(\mu_x) \quad = \quad \beta_0 + \beta_1 x$$

Example 8.2

Suppose D_x is the number of deaths in a year in a group aged x with E_x giving the number exposed to risk, and let q_x denote the mortality rate.

Age x	E_x	d_x/E_x
80	2858	0.110
81	2503	0.122
82	2391	0.133
...		
98	171	0.345
99	124	0.298

- Here D_x is a non-negative integer bounded by E_x, so the simplest model is

$$D_x \quad \sim \quad \text{Binomial}(E_x,\, q_x)$$

  where the D_x are independent.
- Crude mortality estimates are given by

$$\hat{q}_x \quad = \quad \frac{D_x}{E_x}$$

  and notice that $\mathrm{E}\left[\hat{q}_x\right] = q_x$
- We want to model the mortality rate q_x as a function of age x

- A function of the form

$$\log(q_x) = \beta_0 + \beta_1 x$$

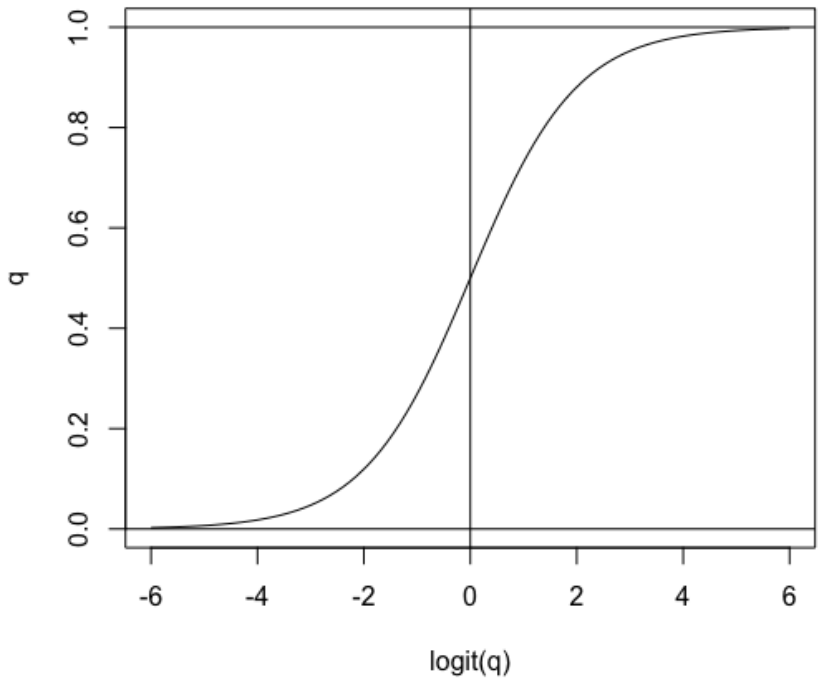
  would only guarantee positive q_x
- We want $q_x \in [0, 1]$
  One option is to use the *logistic transformation*

$$\text{logit}(q_x) \quad = \quad \log\left(\frac{q_x}{1 - q_x}\right) \quad = \quad \beta_0 + \beta_1 x$$

Exercise: Show that if $\text{logit}(q) = \eta$, then

$$q \quad = \quad \frac{\mathrm{e}^{\eta}}{1 + \mathrm{e}^{\eta}}$$

The (inverse) logistic function



'Sigmoid' shape.

Logistic function as log-odds

Consider the odds for death (vs. survival):

$$\frac{P(\text{death})}{P(\text{survival})} = \frac{q_x}{1-q_x}$$

In terms of the logarithm of the odds,

$$\log\left(\frac{P(\text{death})}{P(\text{survival})}\right) = \log\left(\frac{q_x}{1-q_x}\right)$$

So the logistic function applied to the death probability q_x gives the log odds for death.

Logistic GLM

Suppose D_x is the number of successes in E_x trials, and we assume

$$D_x \quad \sim \quad \text{Binomial}(E_x, q_x)$$

where the D_x are independent.

Suppose the trend can be described through the logistic relationship

$$\log\left(\frac{q_x}{1-q_x}\right) \quad = \quad \beta_0 + \beta_1 x$$

This is the *logistic GLM*.